SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

MV Switchgear Panel 12kV (SWGR-MV-12kV) — Datasheet

1. Overview

12kV air-insulated metal-clad switchgear, withdrawable VCB. Product family: MV breakers & switchgear. Lifecycle status: **ACTIVE**.

2. Technical specifications

2.1. Ratings

Item	Value
Rated voltage	12 kV
Rated busbar current	1250 A (up to 2500 A)

2.2. Performance and characteristics

Item	Value
Rated short-time withstand	25 kA / 3 s
Internal arc classification	IAC AFLR 25 kA 1 s
Loss of service continuity	LSC2B-PM
Power frequency withstand	28 kV / 1 min
Lightning impulse withstand	75 kV

3. Installation data

3.1. Terminals

- Dimensions (W x H x D): 800 x 1500 x 2300 (per panel) mm
- Weight: 1100 kg
- Degree of protection: IP4X
- Operating temperature: -5 to +40 °C

4. Standards and certifications

Applicable standards:

- IEC 62271-200

Certifications held:

- KEMA type test report
- CE

Certificates and declarations of conformity are available from the Certification & Compliance team on request. Certifications not listed above are not held for this product; project-specific evaluation can be requested.

5. Ordering information

Item	Value
Ordering code	SWGR-MV-12kV
Standard lead time	14 weeks ex-works
Minimum order quantity	1
Warranty	24 months (see POL-WARRANTY)
Firmware	not applicable

6. Related documents

- IG-SWGR-MV-12kV — Installation and Commissioning Guide
- TG-SWGR-MV-12kV — Troubleshooting and Error Code Guide
- MG-SWGR-MV-12kV — Maintenance Guide
- FAQ-MV_EQUIP — MV breakers & switchgear FAQ

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MV Switchgear Panel 12kV (SWGR-MV-12kV) — Installation and Commissioning Guide

1. Safety

Installation, wiring and commissioning must be performed by qualified personnel. Disconnect and lock out all power sources before work. Verify absence of voltage.

2. Receiving and storage

Inspect the packaging on arrival and report damage within 7 days (see POL-RETURN). Store indoors in a dry place between -25 and +55 °C, relative humidity below 85% without condensation.

3. Mounting and clearances

- Required clearances: 800 mm in front for withdrawal of the circuit breaker truck; 100 mm at the rear unless rear-access type
- Operating temperature: -5 to +40 °C
- Degree of protection: IP4X
- Above 1,000 m altitude, apply derating per the datasheet.

4. Wiring and tightening torques

- Terminal tightening torque: see dimension drawing
- Use copper conductors rated for at least 75 °C.
- Separate power and signal cables by at least 200 mm.

5. Commissioning procedure

1. Check interlocks (circuit breaker, earthing switch, doors)
2. Perform power-frequency withstand test of the main circuit
3. Check protection relay settings by secondary injection
4. Verify CT polarity and VT ratio
5. Rack in and test closing and opening

6. Pre-energisation checklist

☐ Product rating matches the drawing
☐ All terminals tightened to specified torque
☐ Protective earth connected
☐ Insulation resistance measured and recorded
☐ Settings / parameters recorded

7. Common commissioning issues

Symptom	Likely cause	See
Partial discharge alarm in switchgear	Condensation due to anti-condensation heater off	TG-SWGR-MV-12kV §3.1
Partial discharge at cable compartment	Cable termination workmanship defect	TG-SWGR-MV-12kV §3.2
Cable compartment door cannot be opened	Earthing switch not closed (interlock)	TG-SWGR-MV-12kV §3.3
Busbar joint overheating	Insufficient joint torque	TG-SWGR-MV-12kV §3.4

On-site commissioning support can be requested through the Global CS Center. For mv breakers & switchgear the request is handled by Field Service.

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MV Switchgear Panel 12kV (SWGR-MV-12kV) — Maintenance Guide

1. Maintenance intervals

Task	Interval
Mechanism lubrication	5 years or 2,000 operations
Contact wear check	5 years
Thermographic inspection	12 months
Partial discharge check	12 months

2. Spare parts

- No user-replaceable parts; replace the complete unit.

3. End of life

Lifecycle status: ACTIVE. Expected service life is 20 years under normal conditions. Dispose of according to local regulations for electrical equipment.